

Data Intelligence Platform - Data

INTRODUCTION TO DATABRICKS



Kevin Barlow

Data Analytics Practitioner

Why do organizations care about data management?

Protection and security



Confidence in data



Kinds of data

Structured

- Most common and understood
- Typical rows and columns
- **Examples:**
 - database tables
 - .csv
 - Parquet
 - Delta

id	name	occupation	location
1	Kevin	Data Scientist	California
2	Tom	Architect	Arizona
3	Sally	Lawyer	Texas
4	Tina	Surgeon	Florida
5	Joe	Engineer	New York

Kinds of data

Semi-structured

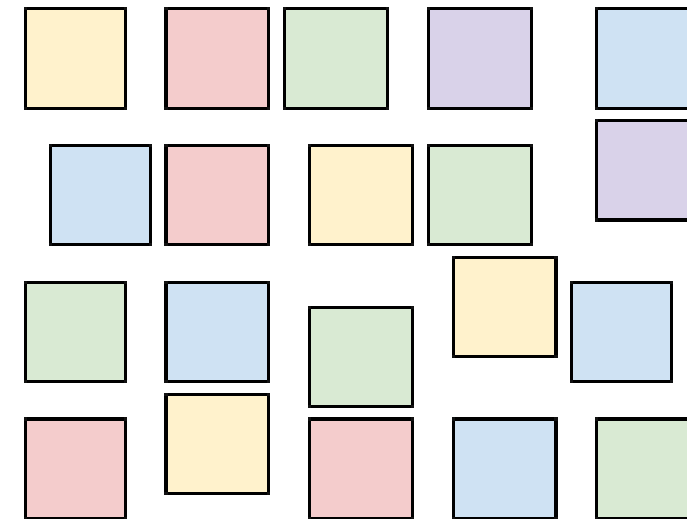
- Common with web-based devices
- Some structure, but more flexible in content
- **Examples:**
 - JSON
 - XML
 - HTML

```
{
  "people": [{
    "id": 1,
    "name": "Kevin",
    "occupation": "Data Scientist",
    "location": "California"},
    {
      "id": 2,
      "name": "Tom",
      "occupation": "Architect",
      "location": "Arizona"}]
}
```

Kinds of data

Unstructured

- Common with smart devices, cameras, etc.
- Little structure, information-rich
- **Examples:**
 - JPEG
 - PNG
 - MP4
 - PDF
 - DOC



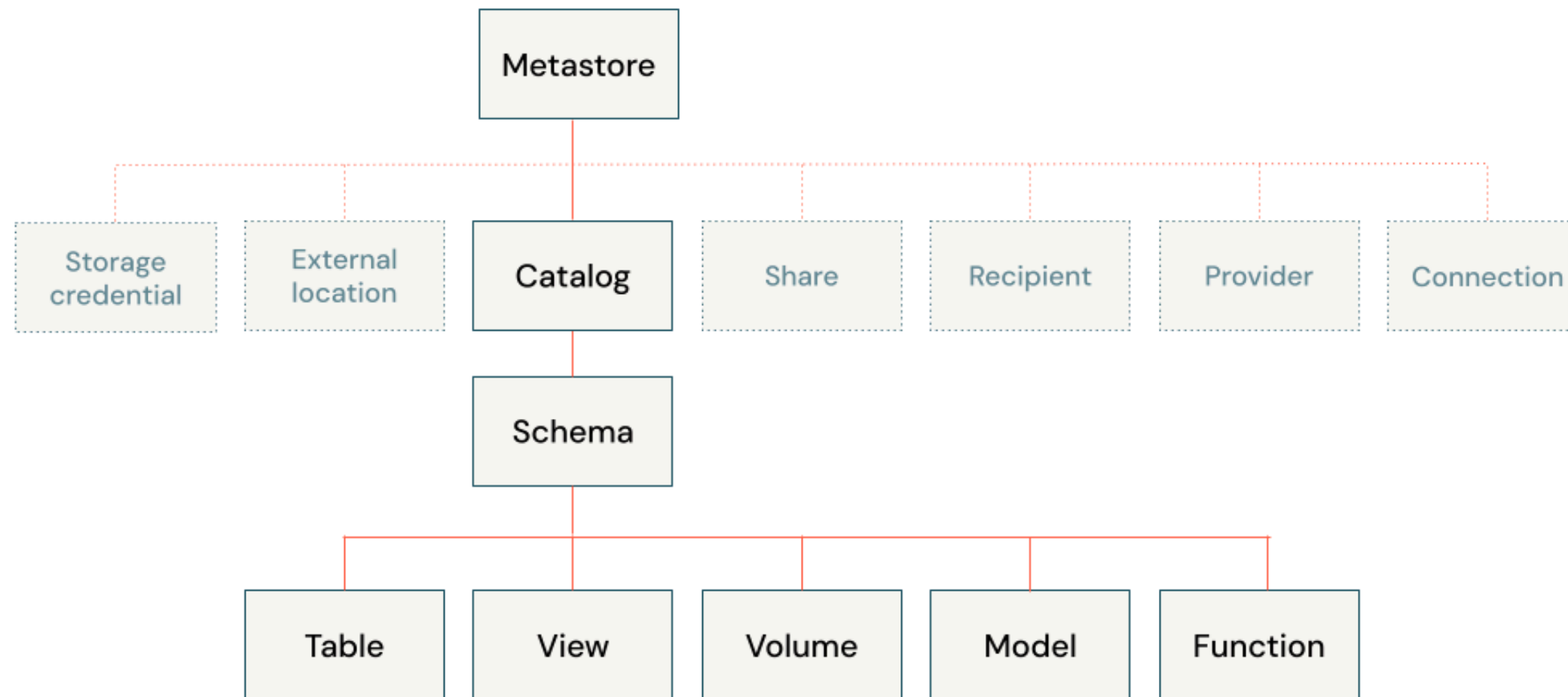
Delta

delta.io

- Open-source storage format
- Collection of parquet tables
- JSON transaction log
- Fully ACID compliant
- Batch and streaming datasets

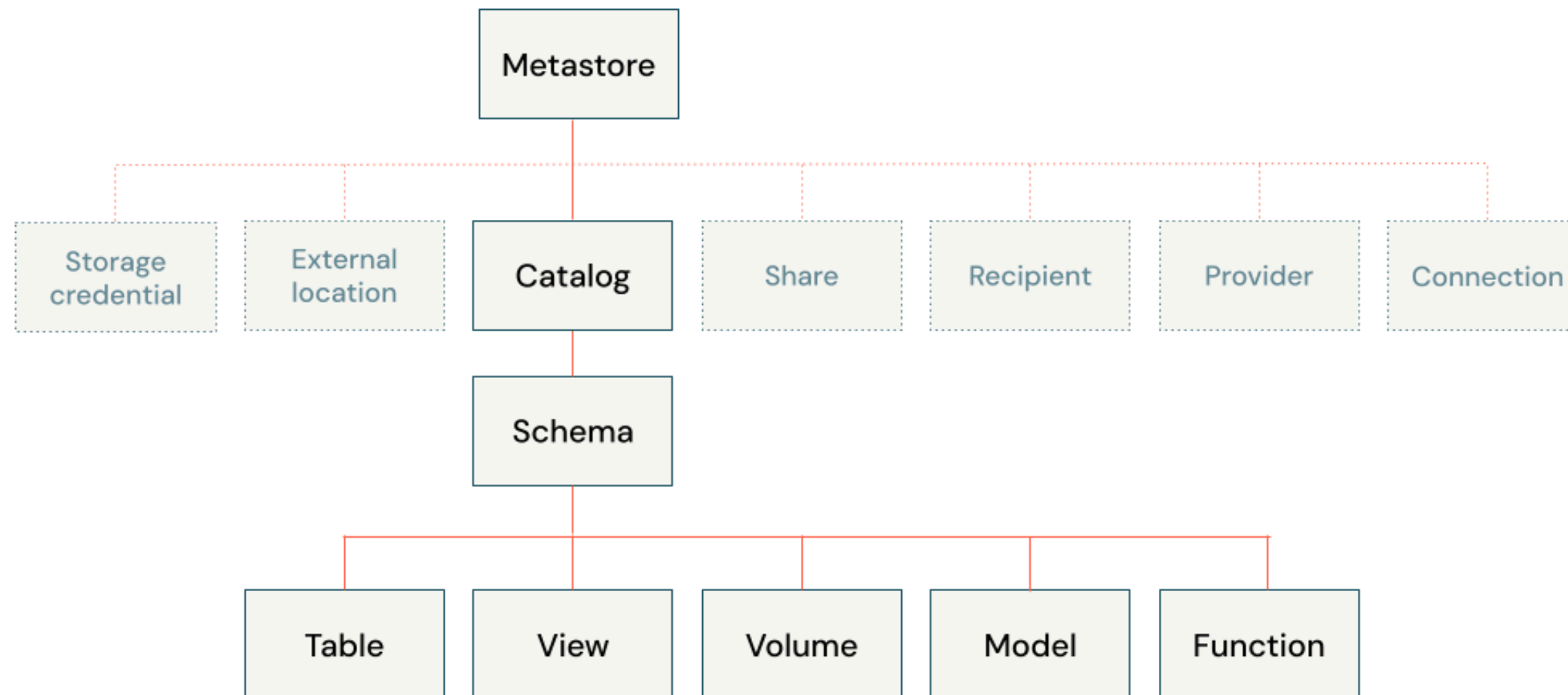


Unity Catalog



¹ <https://docs.databricks.com/en/data-governance/unity-catalog/index.html#the-unity-catalog-object-model>

Unity Catalog



GRANT, SHOW, REVOKE, USE ...

Catalog Explorer

- Single location to explore all data assets
- UI to discover data
- Manage Unity Catalog permissions
- View data lineage and related assets

Catalog Explorer

 Send feedback

+ Add

 Starter Warehouse Pro

2XS 

Catalog



Type to filter



- ✓  databricks_ws_094b2e73_d2f4_4e66_9dd7_4e7a89942f2f
 - >  default
 - >  information_schema
 - >  hive_metastore
- ✓  samples
 - >  default
 - >  nyctaxi
 - >  tpch
- ✓  system
 - >  information_schema

Catalogs

Create catalog

 Filter catalogs

4 catalogs

Name	Owner	Created at
databricks_ws_094b2e73_d2f4_4e66_9dd7_4e7a89942f2f	_workspace_admins_databricks_ws_094b2e73_...	2024-03-21 15:40:44
hive_metastore		
samples		
system	System user	2024-02-01 01:25:41

Let's practice!
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Managing Data Catalogs

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Data Intelligence Platform - Compute

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Why do organizations care about compute?



Apache Spark

- Created by Databricks co-founders
- Open source framework
- Highly efficient distributed computing
- APIs for Python, SQL, Scala, R
- Great for all use cases:
 - data engineering to machine learning and business intelligence

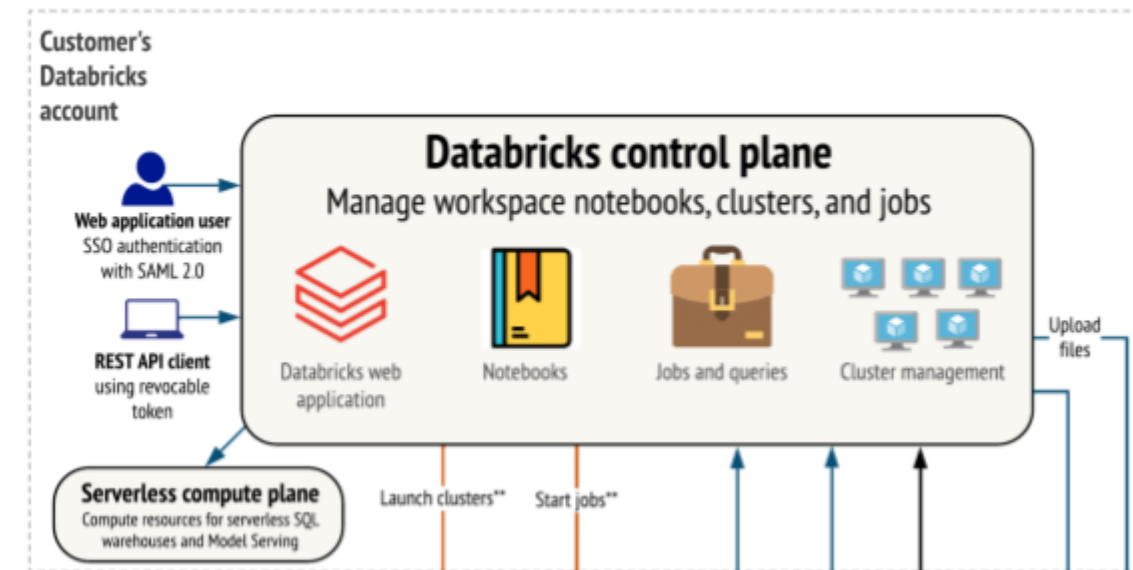
Check out some of the [Apache Spark courses](#) on DataCamp!



Cluster Types

Classic

- Compute resources (virtual machines) are created in the Compute Plane
- Databricks provides configuration to your cloud
- **Pros:** compute and security in your environment, leverage pre-existing compute pools, etc.
- **Cons:** slow startup time

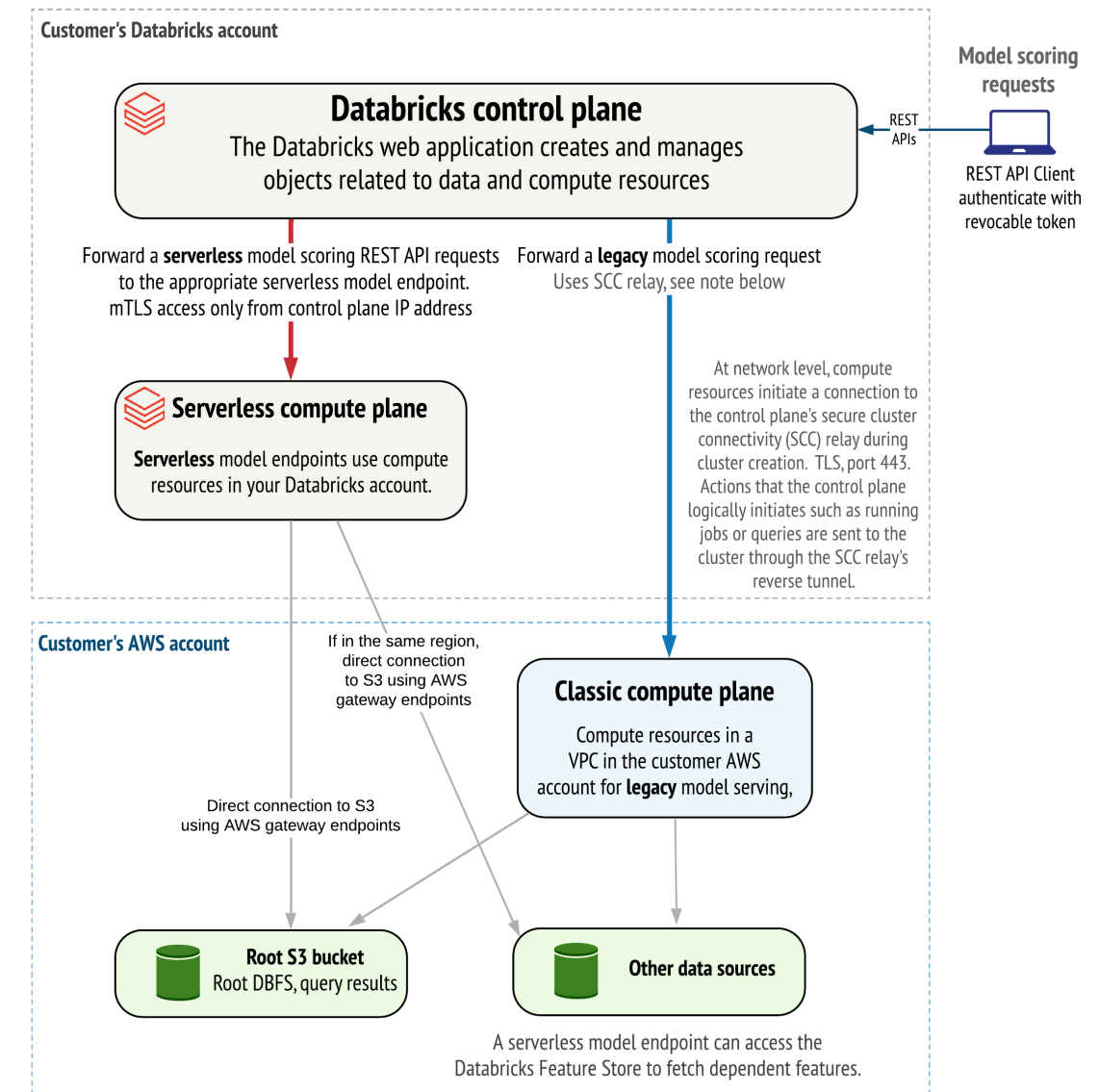


Cluster Types

Serverless

- Compute resources (virtual machines) are created in the Control Plane
- Databricks provides access to your users
- **Pros:** Fast startup time, the latest and greatest feature, the fastest performance, Databricks improves performance over time
- **Cons(?):** compute not in your environment

Compare classic and serverless compute planes for Model Serving



Single-node vs. Multi-node

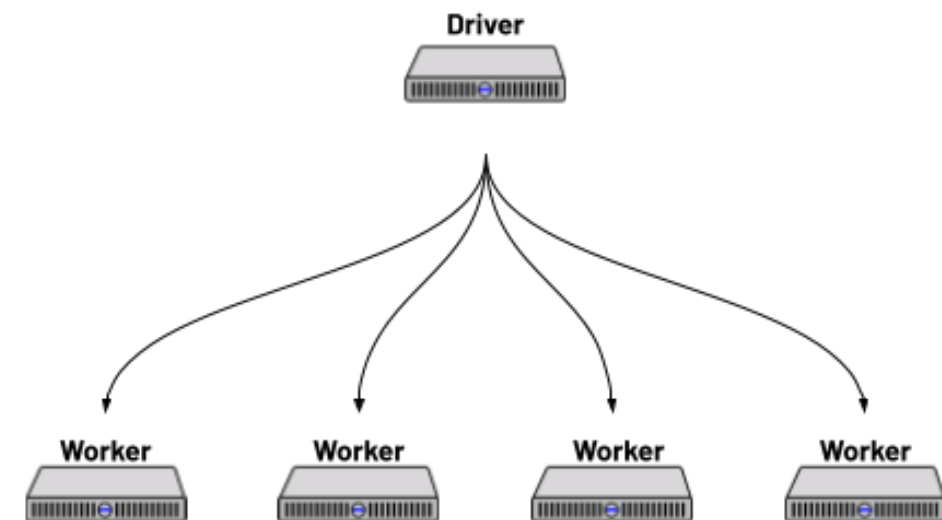
Single-node

- Cluster with just a Driver Node
- Can still run Spark
- Can also run single-node frameworks (i.e., pandas)
- Great for smaller datasets



Multi-node

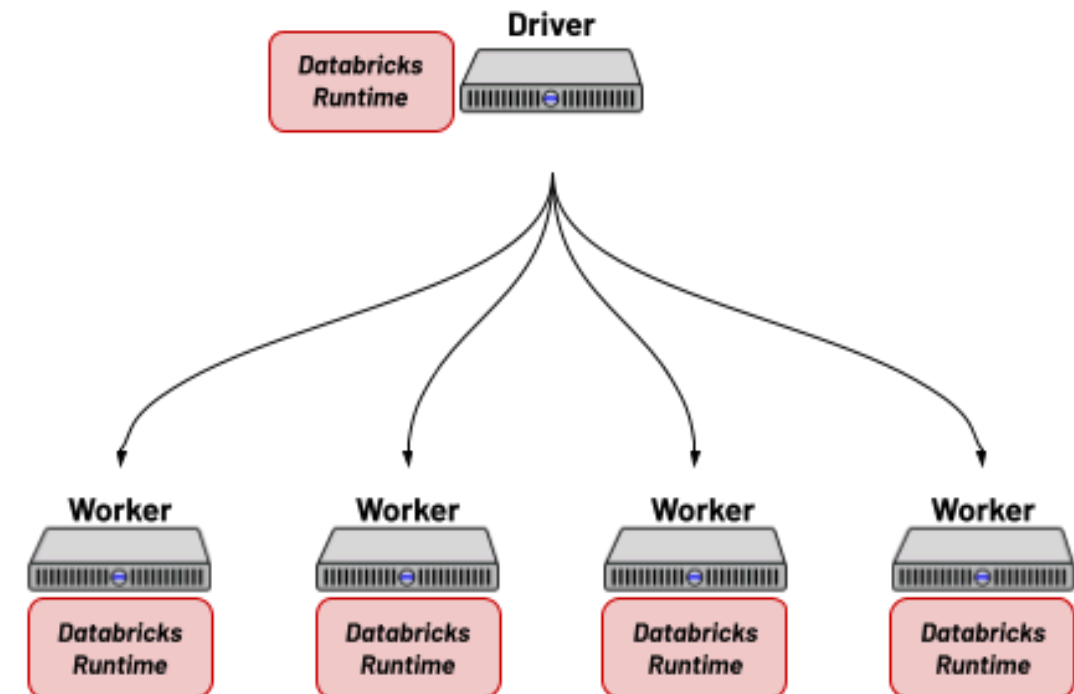
- Cluster with a Driver Node and one or more Worker Nodes
- Spark can distribute work across multiple nodes
- Great for larger datasets



Databricks Runtime

- Installed on every Databricks cluster
 - Optimized version of Apache Spark
 - Photon for faster SQL queries
 - Common libraries (e.g., pandas, dplyr, sci-kit learn)
 - Logic to connect with Databricks services

General recommendation: Use the most recent Long Term Support (LTS) version of the Runtime



Let's practice!
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